

THE PRIME-WAVE ANATOMY OF THE GAP-AMPLITUDE LAW FOR RIEMANN ZEROS

UĞUR SEZEN

ABSTRACT. In a companion note we measured the joint law of consecutive-zero gaps δ_n and between-zero maxima $M_n = \max |Z(t)|$ of Hardy’s function, finding that its correlation matches CUE characteristic polynomials only at a shifted effective dimension, with different shifts for different statistics. Here we dissect the deviation. Regressing $\log M_n$ and $\log \delta_n$ on the explicit-formula waves $\cos(t \log p^k)$ (phases evaluated at gap midpoints, exogenous to the amplitude measurement; placebo controls at the noise floor), we find: (i) a *sum rule* — the total prime content of the amplitudes tracks the explicit-formula weights $p^{-k/2}/k$ term by term across a factor 20 in weight, at an overall normalization of 0.95–1.1 whose exact value is frame-dependent at the 5–15% level; (ii) the content splits between a direct amplitude channel w and a zero-displacement channel v , mirror curves that each collapse onto one-variable laws in $\tau = \log p / \log \frac{t}{2\pi}$; (iii) stripping the measured waves from both series leaves a core pair with correlation $r^* \approx 0.97$ – 0.98 — tighter than any CUE in the sampled range $N \in [5, 20]$ — with the Pearson/Spearman effective-dimension anomaly of the companion note melting away entirely; (iv) an exhaustion analysis reveals a role-crossover at $\tau^* \approx 0.40$ and yields an upper bound on the intrinsic (non-arithmetic) variance of the amplitude sequence, $V_{\text{res}}(L)$, which grows at a rate close to the CUE rate but at one sixth its magnitude. All of these laws, fitted at $t \leq 1.6 \times 10^6$, are then re-tested out of sample at the 10^{12} -th zero ($t \approx 2.7 \times 10^{11}$) using Odlyzko’s tables: the τ -laws, the sum rule, the crossover, and the V_{res} extension all hold across this 1.6×10^5 -fold jump in height; the zero-displacement law is further verified at the 10^{21} -st and 10^{22} -nd zeros, where only gap data exist, extending its validated range to eighteen orders of magnitude in t . At single-gap scale, two thirds or more of the apparent randomness of the maxima sequence is deterministic prime signal.

1. INTRODUCTION

Let γ_n denote consecutive ordinates of nontrivial zeros of ζ , $\delta_n = \gamma_{n+1} - \gamma_n$, $M_n = \max_{\gamma_n \leq t \leq \gamma_{n+1}} |Z(t)|$, and $L = \log \frac{t}{2\pi}$. In a companion note [9] we measured the correlation of the unfolded pair $(\delta_n, \tilde{M}_n)$ over 3.4×10^5 gaps up to $t = 1.6 \times 10^6$, validated the dataset against the proven mean-value results of Conrey–Ghosh [2] and Hughes–Lugmayer–Pearce–Crump [6], and compared it with the CUE characteristic-polynomial model [7]. The outcome was an effective-dimension description with a metric-dependent shift: the rank-based (Spearman) effective dimension is constant, $N_{\text{eff}} = L + 2.0$, while the Pearson-based one drifts from $L + 0.75$ to $L + 1.30$ over the measured range, the two differing by 0.93 ± 0.10 throughout — so no single finite CUE reproduces the joint law.

This note asks *what, concretely, the deviation is*, and answers with a set of regression measurements against the explicit-formula frequencies. The toolbox is elementary (ordinary least squares on 10^5 -size samples), but three design choices matter: all wave phases are evaluated at gap midpoints, which are determined by the zeros alone and are therefore exogenous to the amplitude measurements; every stripping experiment is accompanied by a

placebo run with an equal number of non-prime-power frequencies; and no closed forms are fitted — all comparisons are against measured or proven quantities.

2. DATA AND CONVENTIONS

The dataset is that of [9]: six windows of 4×10^4 gaps centered at $t = 1.2, 2.0, 3.5, 6.0, 10, 16 \times 10^5$ ($L = 9.86$ to 12.45), computed by a vectorized Riemann–Siegel engine validated to 7.6×10^{-6} pointwise and cross-checked against the proven mean of M_n^2 (slope within 0.05%, constant term within 1%). Unfolding uses known scales:

$$\tilde{\delta} = \delta \cdot \frac{L}{2\pi}, \quad \tilde{M} = \frac{M}{\sqrt{AL + \alpha_{-1} + 2A + (\alpha_0 + \alpha_{-1})/L}},$$

with $A = (e^2 - 5)/2$ and the α_i of [6].

For a set $\mathcal{Q}$ of prime powers we regress

$$\log \tilde{M}_n \sim 1 + \sum_{q \in \mathcal{Q}} [a_q \cos(t_n \log q) + b_q \sin(t_n \log q)], \quad \log \tilde{\delta}_n \sim (\text{same basis}),$$

with t_n the midpoint of gap n . “Stripping” means subtracting the fitted wave part from both series and re-exponentiating. Placebo runs replace $\mathcal{Q}$ by an equal number of frequencies bounded away from *all* prime-power lines in the drawing range. The measured floors (worst case over windows and five draws): $|\Delta r| \leq 6 \times 10^{-4}$ and stripped variance $\leq 4 \times 10^{-4}$ for the 9-frequency basis, growing to $|\Delta r| \leq 5 \times 10^{-3}$ and variance $\leq 2 \times 10^{-3}$ for the largest (79-frequency) basis. All quoted effects exceed these floors by an order of magnitude or more, except where noted.

3. THE SUM RULE

The explicit formula assigns the wave $\cos(t \log p^k)$ weight $p^{-k/2}/k$ in $\log |\zeta(\frac{1}{2} + it)|$. Define the normalized content $u_q = \sqrt{a_q^2 + b_q^2} / (p^{-k/2}/k)$ for $q = p^k$.

Observation 1 (Sum rule). *For all 23 frequencies tested — the primes 2 through 47 and the prime powers 4, 8, 9, 16, 25, 27, 32, 49 — the measured total content of the maxima tracks the explicit-formula weight, which itself spans a factor of 20 (0.71 down to 0.035). In the midpoint frame $|u_q| = 1.01$ to 1.12 . The sine components vanish (phase-aligned waves), and placebo frequencies give $|u| \approx 0$.*

The overall normalization carries a *frame systematic*: the gap midpoints are themselves displaced by the prime waves (the v channel of the next section), so they carry phase jitter (rms ≈ 0.14 , a quarter of the mean gap). Re-evaluating the phases at jitter-free positions — the Riemann–von Mangoldt flow started at the window’s first zero — moves every prime to $|u_p| = 0.95$ – 0.97 with *no* frequency drift (the apparent +1% to +12% drift of the midpoint frame is a frame artifact), while the low-weight prime powers become noisier but remain consistent with unity. We therefore state the sum rule as: term-by-term proportionality to $p^{-k/2}/k$ is frame-robust; the overall level is ≈ 1 within a 5–15% frame systematic, the jitter-free frame preferring ≈ 0.96 — intriguingly close to the $\tau \rightarrow 0$ transmission limit measured below.

4. TWO CHANNELS AND THE τ -LAWS

The unit total splits into two measurable channels: a direct amplitude channel (coefficient w_p : the response of $\log \tilde{M}$ at fixed observed gap) and a zero-displacement channel (coefficient v_p : the response of $\log \tilde{\delta}$). Measured across the six windows, the two run in opposite directions:

p	2	3	5	7	11	13
w_p	0.76	0.65	0.50	0.41	0.30	0.26
$ v_p $	0.13	0.20	0.29	0.35	0.42	0.45

and both collapse, across all windows and primes simultaneously, onto single curves in the Berry variable $\tau = \log p/L$ [1] (residual RMS ≈ 0.01 for w). The phenomenon underlying the v channel — non-uniformity of the zero ordinates against the prime frequencies — is classical: Ford and Zaharescu [3] (see also [4]) proved that the fractional parts $\{\gamma \log p/2\pi\}$ have a non-uniform limiting distribution tied to the primes. What is new here is the measured per-gap transmission coefficient and its one-variable τ -law. Long waves (small τ) transmit into amplitude; short waves are absorbed into zero positions. The channels are roughly consistent with the sum rule through the gap–amplitude coupling: $w_p + \beta v_p$ lies within 5–30% of unity (the discrepancy growing with p), with $\beta \approx 2.3$ the log-log regression slope of amplitude on gap. We read this as a measured, observable-level decomposition of the hybrid Euler–Hadamard product of Gonek–Hughes–Keating [5]: the “zeros part” and “primes part” are here separated channel by channel, with the partition governed by τ alone.

5. STRIPPING: THE ARITHMETIC CORE

Removing the fitted waves from both series (basis: all $p^k \leq Q$) and recomputing the correlation of the residual pair gives, at $Q = 49$, $r_P^* \approx 0.97$, $r_S^* \approx 0.98$ — far tighter than any CUE in the range $N \in [5, 19]$ (max 0.93) — while the placebo leaves the raw values 0.75/0.88 untouched. Two structural consequences follow.

Observation 2 (The effective-dimension anomaly is arithmetic). *After stripping, the Pearson and Spearman statistics of the residual pair coincide (both 0.9734 at $L = 11.47$; raw discrepancy 0.12 shrinks to 0.008). The two-metric anomaly of [9] is entirely a prime-wave effect.*

Observation 3 (Exhaustion curve). *Extending the basis to all $p^k \leq Q$ with Q up to 300 (79 frequencies; corrected placebo floor $|\Delta r| \leq 5 \times 10^{-3}$ for this basis size), the stripped correlation $r^*(Q)$ rises to a maximum and then declines. The post-peak decline is largest in the lowest windows (-0.013 at $L = 9.86$, several times the floor) and shrinks with L (-0.0013 at $L = 12.45$, within the floor); the peak location is therefore anchored by the lower windows. The maximum sits at $\log Q_{\text{peak}}/L \approx 0.40 \pm 0.03$ across windows (the Q grid is coarse): below this threshold the waves act as amplitude noise (stripping tightens the pair), above it as common-mode signal shared by gaps and maxima (stripping removes genuine coupling). We denote this role-crossover $\tau^* \approx 0.40$.*

At $Q = 300$ the stripped share of the amplitude log-variance reaches 82–87%, leaving a residual

$$V_{\text{res}}(L) : \quad 0.163 \ (L=9.86) \rightarrow 0.244 \ (L=12.45),$$

increasing approximately linearly with slope ≈ 0.031 per unit L . For comparison, the CUE amplitude log-variance grows at ≈ 0.036 per unit N but is six times larger in magnitude at

matched dimension. V_{res} is an *upper bound* on the intrinsic randomness of the maxima sequence: our per-frequency coefficients are gap-independent, whereas the conditional-variance anatomy shows the transmission grows with gap width, so arithmetic residue survives the stripping.

The $\tau \rightarrow 0$ limit. A dedicated small- τ campaign — four additional windows at $t = 10^8, 10^9, 10^{10}, 10^{11}$ (2×10^4 – 3×10^4 gaps each; the 10^{11} window computed with an anchor-expanded engine after a double-precision phase-noise artifact was caught and eliminated by a placebo-style consistency check against the Odlyzko window) — yields a 44-point table of $w(p, L)$ reaching $\tau_2 = 0.028$. Two findings. First, the τ -collapse acquires a resolvable fine structure: at fixed τ , larger primes transmit slightly less (e.g. at $\tau \approx 0.058$, $w_2 = 0.771$ versus $w_3 = 0.753$, a 4σ split); a secondary variable beyond τ exists, plausibly related to the u_q drift noted above. Second, extrapolating $\tau \rightarrow 0$ under five model variants (pure quadratic; quadratic plus $1/p$; plus $1/L$; $p = 2$ points only; extended prime set) gives $w(0)$ between 0.88 and 0.98, with *every* variant below unity, though at varying strength — the cleanest single-prime variant ($p = 2$ only) is compatible with unity at 1.5σ , while the multi-prime variants exclude it at 4.7σ or more:

Observation 4 (Incomplete transparency). *The amplitude-channel transmission does not reach unity as $\tau \rightarrow 0$: a persistent absorption of 2–12% survives even for the longest prime waves. The exact value is currently blocked by a degeneracy between $1/p$ and $1/L$ corrections (at fixed τ these are tied through $\log p = \tau L$).*

6. OUT-OF-SAMPLE VALIDATION AT THE 10^{12} -TH ZERO

All laws above were extracted from heights $t \leq 1.6 \times 10^6$. Using Odlyzko’s table of the 10^4 zeros following the 10^{12} -th [8] ($t \approx 2.677 \times 10^{11}$, $L = 24.475$), we recomputed the entire battery at a height larger by a factor of 1.6×10^5 . Amplitudes were evaluated by an anchor-expanded Riemann–Siegel engine (206,393-term main sum; base phases $\theta(t_0) - t_0 \log n \bmod 2\pi$ computed once in high precision, increments in double precision); at the tabulated zeros the engine gives median $|Z| = 1.4 \times 10^{-9}$ against between-zero maxima of order 3, validating it to nine digits.

The results, with the small-height laws quoted as predictions:

- *τ -law for w :* the linear guide $0.940 - 2.967\tau$, fitted only to small heights, predicts the six measured w_p at $L = 24.475$ to within 0.02–0.07; e.g. $w_2 = 0.871 \pm 0.007$ against the guide value 0.856. The residuals grow mildly with τ (the true curve is not exactly linear), but the collapse variable survives the jump.
- *Sum rule:* $|u_q| = 0.99$ – 1.07 across the twelve frequencies tested (uncertainties ± 0.02 at $q = 2$ growing to ± 0.2 at the largest q ; the $q = 49$ value 1.18 is 0.9σ from unity).
- *Role-crossover:* at this L the crossover law places $Q_{\text{peak}} = e^{0.40L} \approx 2 \times 10^4$, so the entire accessible range $Q \leq 300$ ($\tau \leq 0.233$) should lie *before* the peak: $r^*(Q)$ should rise without declining. It does ($0.901 \rightarrow 0.959$), in contrast to the peak-and-decline seen at $L = 12.45$ over the same Q range — a successful qualitative out-of-sample prediction.
- *Intrinsic variance:* the linear extension of $V_{\text{res}}(L)$ predicts 0.617 at this height; the measured value is 0.650, high by 5% in the direction expected from the (necessarily) incomplete stripping at this L .

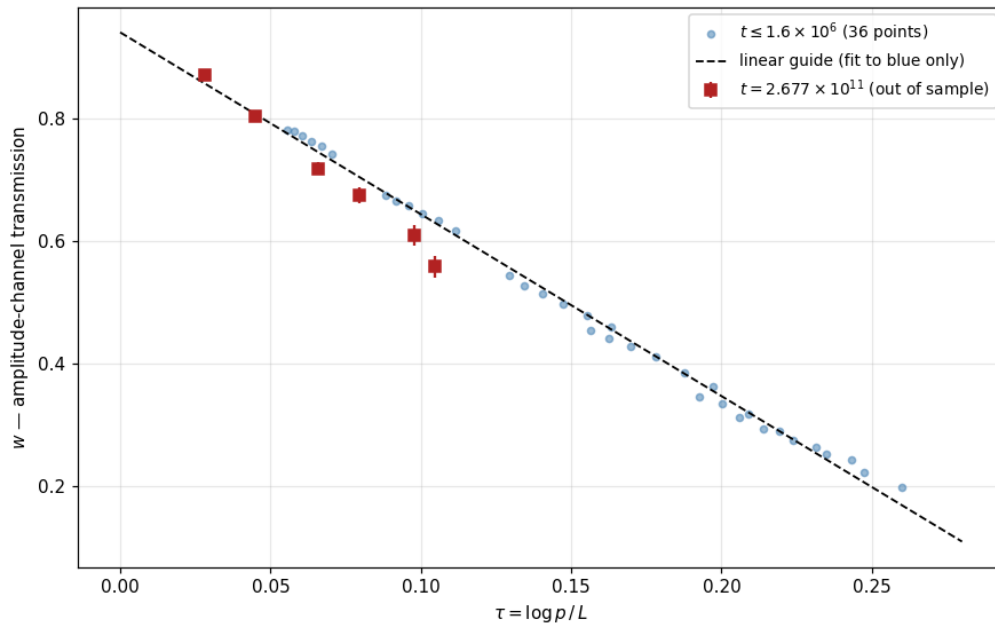


FIGURE 1. The amplitude-channel transmission w against $\tau = \log p/L$. Blue: the 36 measurements at $t \leq 1.6 \times 10^6$. Red: the six out-of-sample measurements at $t = 2.677 \times 10^{11}$. Dashed: the linear guide fitted only to the blue points.

- *Effective dimension*: the N_{eff} test is *inconclusive* here, neither confirming nor refuting the constant shifts: $r_{\text{CUE}}(N)$ flattens at large N ($dr/dN \approx 0.008$), so with 10^4 gaps the uncertainty on N_{eff} is ≈ 0.8 . The raw deficit $r_\zeta - r_{\text{CUE}}(N=L) = -0.014 \pm 0.006$ is consistent with the small-height value -0.019 .

Second-order lines. Prompted by an adversarial re-analysis of our own pipeline, we searched the amplitude series for content at *composite* frequencies $\log(pq)$ and $\log(p/q)$, $p, q \in \{2, 3, 5, 7\}$ — frequencies at which the von Mangoldt function vanishes, so the linear explicit formula predicts nothing. The product lines are present at the $\text{few} \times 10^{-3}$ level: individually $2\text{--}5\sigma$ per window, but coherently *cosine-locked with negative sign* in all 13 significant (window, line) measurements across three windows — a sign pattern with chance probability $\sim 2^{-13}$. The difference lines are much weaker or absent. This asymmetry refutes the simplest mechanism (a quadratic rectification of the real wave sum, which would feed sums and differences equally) and suggests the second-order response is *analytic* in the wave phases, retaining $e^{i(\theta_p + \theta_q)}$ without the conjugate cross-terms. The total log-variance carried by these lines is $\sim 10^{-4}$, negligible for V_{res} ; their interest is structural, not budgetary.

7. THE GAP CHANNEL AT THE 10^{21} -ST AND 10^{22} -ND ZEROS

At heights $t \sim 10^{20}$ the Riemann–Siegel main sum has $\sim 10^{10}$ terms and amplitude computation is out of reach. The zero-displacement channel, however, requires only zero *locations*. Using Odlyzko’s tables of the 10^4 zeros following the 10^{21} -st and 10^{22} -nd zeros [8] ($L = 44.58$ and 46.83 ; stated accuracy 10^{-6} ; wave phases anchored at the tables’ integer base offsets), we regressed the unfolded gap sequence on the six prime waves. All twelve (prime, height) coefficients are detected at $3\text{--}5\sigma$ and agree with the small- τ guide $v = 2.014\tau$ — fitted

only to the $t \leq 2.7 \times 10^{11}$ data — within their uncertainties; the most distant point gives $v_2 = 0.0297 \pm 0.0095$ against the predicted 0.0298. Placebo frequencies sit at the noise floor (Figure 2).

The gap-channel τ -law is thereby verified over *eighteen orders of magnitude* in height: re-measuring the channel on the low-height dataset of [9] extends it down to $t \approx 1.1 \times 10^3$, and the present tables carry it to 1.4×10^{21} . The low windows also reach τ up to 0.46, where the v -curve is seen to bend away from its linear onset and saturate near 0.55–0.60 — in the same region as the role-crossover $\tau^* \approx 0.40$ of the exhaustion analysis, as one would expect if absorption is nearly complete there. We also note the clean linear onset $v \approx 2\tau$ as $\tau \rightarrow 0$, a first coefficient for any future screening theory of the zero gas.

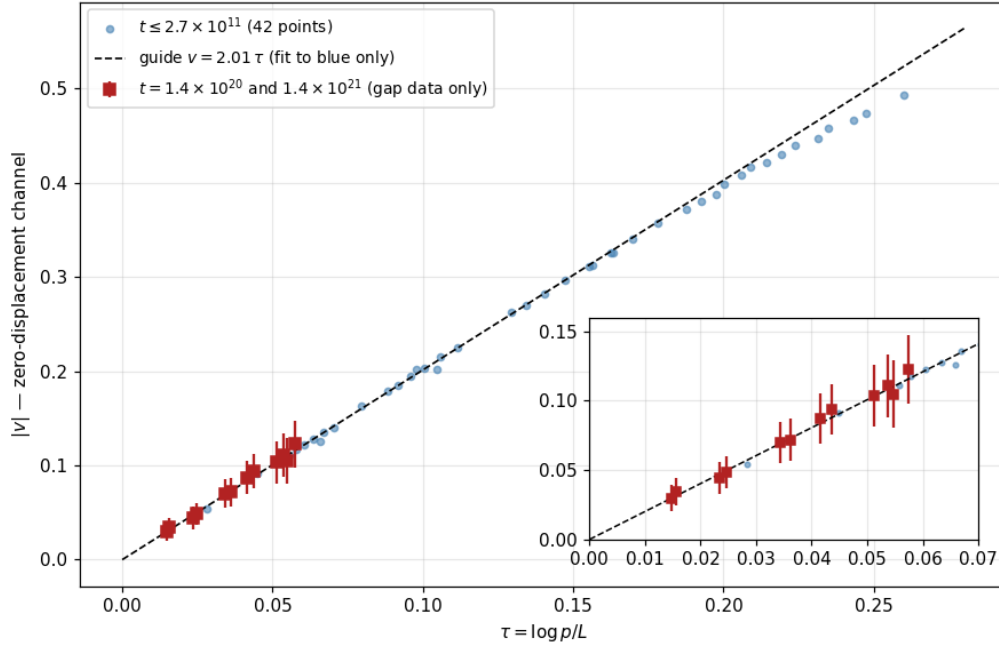


FIGURE 2. The zero-displacement transmission $|v|$ against $\tau = \log p/L$. Blue: measurements at $t \leq 2.7 \times 10^{11}$. Red: the twelve gap-only measurements at $t = 1.4 \times 10^{20}$ and 1.4×10^{21} . Dashed: the linear guide fitted only to the blue points. Inset: the small- τ region.

8. WHAT THIS SAYS, AND WHAT IT DOES NOT

At single-gap scale, the “randomness” of $|Z|$ between zeros is, to two thirds or more, deterministic prime signal carried at full explicit-formula weight; the celebrated CUE resemblance of the gap–amplitude law is a single-statistic compensation, not a model equivalence. None of this bears directly on the Riemann Hypothesis, and none of it is a closed-form claim: the two candidate closed forms this project produced at smaller heights were both refuted by larger- T tests and are recorded as such in the project log. What survives are measured curves — $w(\tau)$, $v(\tau)$, the sum rule $u \approx 1$, the crossover $\tau^* \approx 0.40$, and $V_{\text{res}}(L)$ — each stable across a factor 13 in height within the primary windows (and far beyond in the out-of-sample tests), each protected by placebo controls.

Open problems: (1) derive $w(\tau)$ and $v(\tau)$ — equivalently the partition of the unit explicit-formula weight — from a screening theory of the zero gas; (2) explain $\tau^* \approx 0.40$; (3) determine the nature of V_{res} , whose CUE-like growth rate at reduced magnitude suggests a small genuinely-random component of the zero gas; (4) account for the +1% to +12% drift of u_q and the fixed- τ per-prime splitting, plausibly one phenomenon; (5) break the $1/p$ -versus- $1/L$ degeneracy and pin the residual absorption $1 - w(0) \in [0.02, 0.12]$; (6) recover the discriminating power of the N_{eff} test at large heights, which needs $\gtrsim 10^5$ consecutive zeros there (e.g. from the Platt/LMFDB datasets); (7) explain the sum-only, sign-coherent structure of the second-order lines.

REPRODUCIBILITY

All measurements are scripts 27–60 in <https://github.com/azizran/deney> (directory `qm_riemann/`), runnable from the stored gap–maximum datasets; the dated research log in the same directory records the failed intermediate models (independent-phase and common-phase generative attempts) alongside the results reported here.

ACKNOWLEDGEMENTS

Computational assistance, literature search and drafting: Claude (Anthropic). Errors are the author’s.

REFERENCES

- [1] M. V. Berry, *Semiclassical formula for the number variance of the Riemann zeros*, Nonlinearity **1** (1988), 399–407.
- [2] J. B. Conrey, A. Ghosh, *A mean value theorem for the Riemann zeta-function at its relative extrema on the critical line*, J. London Math. Soc. (2) **32** (1985), 193–202.
- [3] K. Ford, A. Zaharescu, *On the distribution of imaginary parts of zeros of the Riemann zeta function*, J. reine angew. Math. **579** (2005), 145–158.
- [4] K. Ford, K. Soundararajan, A. Zaharescu, *On the distribution of imaginary parts of zeros of the Riemann zeta function, II*, Math. Ann. **343** (2009), 487–505.
- [5] S. M. Gonek, C. P. Hughes, J. P. Keating, *A hybrid Euler–Hadamard product for the Riemann zeta function*, Duke Math. J. **136** (2007), 507–549.
- [6] C. Hughes, S. Lugmayer, A. Pearce-Crump, *The second moment of the Riemann zeta function at its local extrema*, arXiv:2411.05573 (2024); J. London Math. Soc. (2) (2025), e70250.
- [7] J. P. Keating, N. C. Snaith, *Random matrix theory and $\zeta(1/2 + it)$* , Comm. Math. Phys. **214** (2000), 57–89.
- [8] A. M. Odlyzko, *Tables of zeros of the Riemann zeta function*, https://www-users.cse.umn.edu/~odlyzko/zeta_tables/.
- [9] U. Sezen, *The joint law of zero gaps and local maxima of Hardy’s Z-function: a numerical study and a CUE effective-dimension anomaly*, draft (2026), same repository.

INDEPENDENT RESEARCHER, ISTANBUL, TURKEY
 Email address: ugursezen@gmail.com